

VINH, Viet Nam

WMO: 488450

Lat: 18.667N

Lon: 105.683E

Elev: 20

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	52.4	54.3	45.5	45.1	57.5	48.4	50.2	58.0	11.9	68.1	10.0	67.2	2.0	320	0.263

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	11.6	99.0	80.6	97.1	80.3	95.2	80.0	83.2	92.0	82.4	90.7	81.8	89.8	7.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.2	162.2	87.2	80.5	158.2	86.6	79.8	154.9	85.9	47.3	92.3	46.4	90.9	45.7	90.2	93.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.3	10.8	9.1	DB	49.6	101.5	2.8	1.6	47.6	102.7	45.9	103.7	44.4	104.6	42.4	105.8	
			WB	47.6	86.0	2.8	3.0	45.6	88.2	43.9	90.0	42.3	91.7	40.2	93.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	77.1	64.7	66.4	70.6	78.0	84.1	87.7	86.9	85.2	82.3	78.3	73.2	66.9
	DBStd	9.42	5.90	6.81	5.85	5.74	4.97	3.77	3.59	3.13	3.46	3.55	4.86	5.29
	HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0
	HDD65	202	80	61	16	1	0	0	0	0	0	0	3	41
	CDD50	9882	458	460	637	839	1056	1131	1145	1092	968	878	695	523
	CDD65	4609	72	101	189	390	591	681	680	627	518	413	248	99
	CDH74	48100	94	277	703	3167	7083	9451	9203	7892	5582	3211	1209	228
	CDH80	21170	7	48	156	1014	3306	5238	4842	3631	2016	716	184	12
(14) Wind	WSAvg	2.9	2.2	2.3	2.4	2.8	3.1	4.1	3.9	3.2	2.6	2.7	2.6	2.4
(15) Precipitation	PrecAvg	78.7	1.9	1.5	2.2	2.5	6.5	4.1	5.3	9.6	17.8	17.8	6.4	3.1
	PrecMax	103.9	4.5	2.7	5.6	6.7	16.7	12.4	14.9	28.2	37.0	47.2	18.2	9.8
	PrecMin	48.7	0.0	0.6	0.6	0.6	1.5	0.2	0.9	2.1	5.3	1.9	1.3	1.1
	PrecStd	14.3	1.1	0.6	1.1	1.4	3.5	3.3	3.0	6.5	8.3	12.2	4.4	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.3	86.5	93.0	99.2	101.1	100.8	100.7	98.5	95.8	89.0	86.6	81.7
		MCWB	72.4	73.7	77.5	80.7	81.6	81.3	80.5	80.5	80.6	78.9	78.6	75.0
	2%	DB	76.4	79.7	83.6	93.2	98.2	98.7	97.8	95.6	92.9	87.1	84.2	78.8
		MCWB	71.1	73.7	76.8	80.6	80.9	80.5	80.1	80.2	80.7	78.0	76.9	73.2
	5%	DB	74.3	77.0	80.1	88.4	95.2	96.7	95.7	93.4	90.4	85.5	82.1	76.5
		MCWB	70.1	72.7	75.3	79.9	80.3	80.2	79.9	79.9	80.1	77.3	75.5	72.1
	10%	DB	72.3	74.9	77.4	85.3	92.1	94.6	93.6	91.4	88.3	84.0	80.0	74.4
		MCWB	69.0	71.8	73.8	78.6	80.4	79.8	79.7	79.8	79.4	76.9	74.5	70.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.7	76.1	79.5	83.0	84.3	84.3	83.7	83.2	83.1	81.0	79.6	76.4
		MCDB	77.6	81.3	86.4	92.9	94.2	93.9	92.2	91.6	90.5	86.1	84.6	79.4
	2%	WB	72.1	74.5	77.1	81.3	82.8	82.8	82.4	82.1	81.9	79.6	78.2	74.6
		MCDB	75.1	77.9	82.2	89.9	91.6	91.6	91.2	89.8	88.5	84.2	82.2	77.6
	5%	WB	70.9	73.4	75.7	80.1	81.9	82.0	81.6	81.5	81.1	78.8	77.1	72.9
		MCDB	73.5	76.5	79.6	87.3	90.2	90.5	90.2	88.9	87.4	83.1	80.6	75.6
	10%	WB	69.6	72.2	74.4	79.0	81.1	81.3	80.8	80.8	80.3	78.1	76.0	71.5
		MCDB	72.0	74.9	77.3	84.7	89.1	89.7	89.3	88.0	86.1	82.1	79.1	74.1
(35) Mean Daily Temperature Range	5% DB	MDBR	7.5	7.5	8.1	9.7	11.5	11.6	11.1	10.7	9.7	8.6	8.7	8.0
		MCDBR	10.1	10.9	11.9	15.1	15.9	14.1	14.0	13.6	13.0	10.9	11.2	10.5
	5% WB	MCWBR	5.0	5.0	5.5	6.0	4.7	4.1	4.1	4.4	4.4	3.6	4.5	5.1
		MCWBR	9.0	10.0	11.6	13.2	13.5	12.6	12.4	11.9	11.5	9.2	10.0	9.2
(40) Clear-Sky Solar Irradiance	taub	0.582	0.619	0.731	0.682	0.532	0.472	0.450	0.491	0.562	0.659	0.597	0.628	0.628
		taud	1.813	1.772	1.591	1.721	2.127	2.310	2.366	2.237	2.008	1.738	1.832	1.726
	Ebn at Noon	222	222	204	216	249	261	268	259	240	210	216	204	204
		Edh at Noon	64	70	87	76	50	42	39	45	56	71	62	68
(44) All-Sky Solar Radiation	RadAvg	804	990	1083	1415	1641	1645	1570	1482	1321	1095	968	751	751
	RadStd	150	149	108	126	147	120	188	113	148	150	132	121	121

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	+0.77	N/A	N/A	+1.55	+0.56	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.59	N/A	N/A	+0.79	+0.36	+0.29	N/A	N/A	N/A	N/A

Nomenclature: See separate page